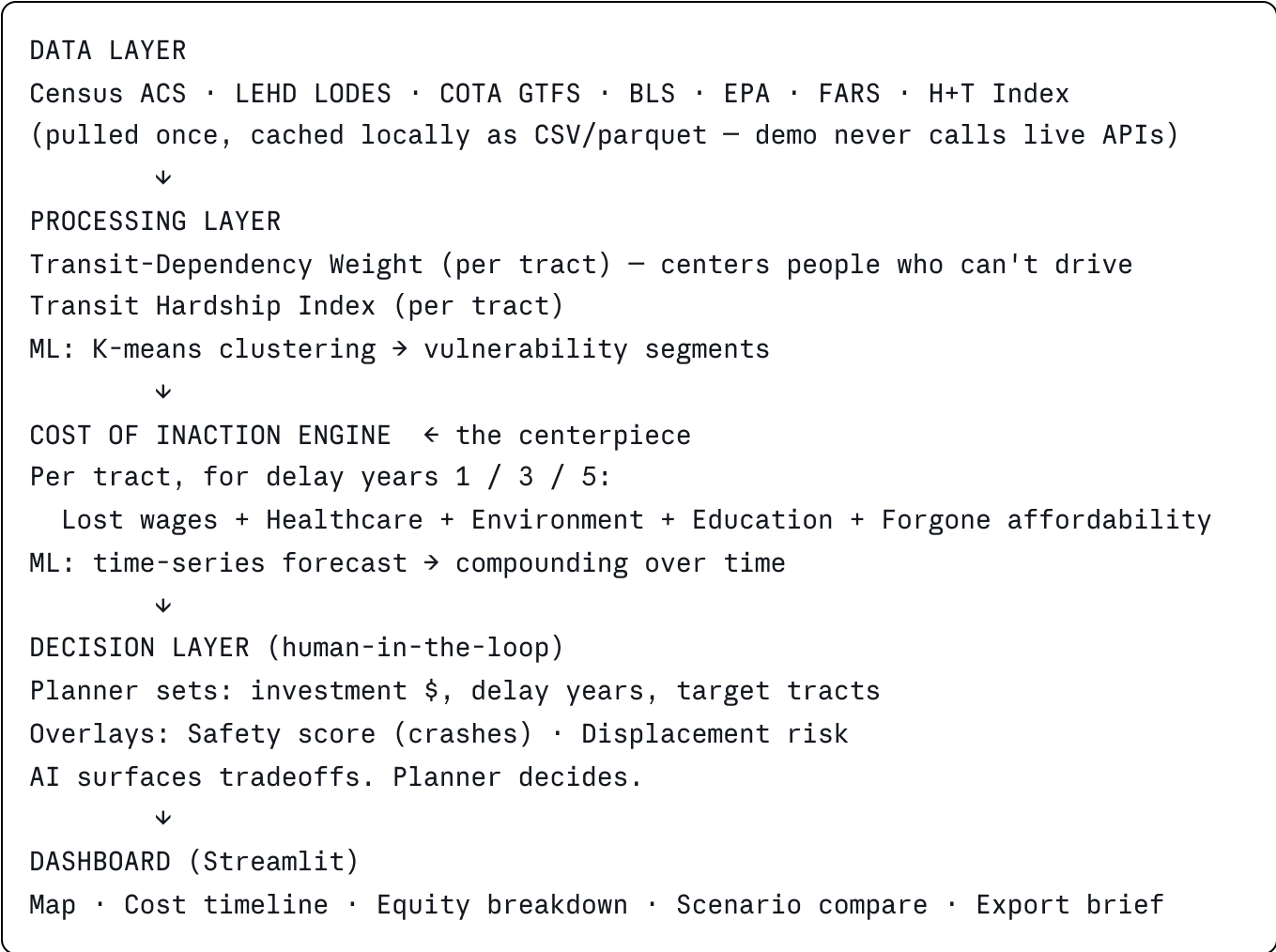


# TransitIQ — System Architecture

**Challenge:** USAII Hackathon 2026, Graduate Track, Brief 6A (Cost of Doing Nothing Simulator) **City:** Columbus, OH **User:** City transit + budget planner deciding "fund now vs. defer" **Core output:** "If this transit investment is delayed N years, here is the compounding cost to low-income communities — in dollars, by neighborhood."

## 1. Architecture (5 layers)



## 2. Data Sources (all public, all free, all tract-level)

| Source     | What it gives                                                               | Access         |
|------------|-----------------------------------------------------------------------------|----------------|
| Census ACS | Income, poverty, vehicle access, commute time, disability, enrollment, rent | API (free key) |

| Source           | What it gives                                      | Access                       |
|------------------|----------------------------------------------------|------------------------------|
| LEHD<br>LODES    | Home→work job locations (origin-destination)       | Flat file download           |
| COTA GTFS        | Columbus bus routes, stops, schedules              | COTA portal /<br>Transitland |
| BLS OEWS         | Wages by occupation (wage cross-check)             | API / flat file              |
| EPA              | Social cost of carbon (\$190/ton), emission factor | Published constants          |
| NHTSA<br>FARS    | Geocoded fatal crashes (safety layer)              | cdan.dot.gov                 |
| CNT H+T<br>Index | Combined housing + transport cost per tract        | Free download                |

### 3. Calculations (the cost engine)

Each component is an **annual cost per tract**. All five sum to the yearly "cost of doing nothing." Compounding across delay years gives the headline number.

#### 3.0 Transit-Dependency Weight (apply first)

Not everyone can substitute a car. Some residents have no alternative to transit — people who can't drive (disability, medical, never learned), older adults who've aged out of driving, and single-parent households. These people bear the real cost of inaction, so the model weights them most heavily instead of averaging all residents equally.

```
transit_dependency_weight = normalized sum of:
  no-vehicle households      (ACS B08201)
+ population with disability (ACS B18101)
+ elderly population (65+)   (ACS B01001 age bands)
+ single-parent households   (ACS B11003)
```

This weight scales the affected-population term in the cost components below, so a tract full of transit-dependent residents registers a higher cost than an equally-sized tract where most people can drive. Rideshare is not treated as a substitute — at \$15-40/trip it is not a viable daily-access option for low-income transit-dependent households.

### 3.1 Lost Wages (time cost)

Method: USDOT Value of Travel Time — local personal travel valued at 50% of wage.

```
lost_wages = excess_commute_hours_per_year
            × hourly_wage
            × 0.50
            × affected_workers
```

- $\text{excess\_commute\_hours} = (\text{tract commute time} - \text{benchmark}) \times 2 \text{ trips} \times 260 \text{ workdays}$
- $\text{hourly\_wage} = \text{tract median income} \div 2,080$
- $\text{affected\_workers} = \text{transit commuters} + \text{no-vehicle workers}$
- Fields: ACS B08303, B19013, B08141, B08301

### 3.2 Healthcare Burden

Method: transit-dependent residents miss care (Urban Institute: 5%–20%).

```
healthcare = transit_dependent_pop
            × missed_care_rate      (0.05 base → 0.20 no-vehicle; slider)
            × cost_per_missed_episode (planner slider)
```

- Fields: ACS B08201, B18101, C27007

### 3.3 Environmental Cost

Method: forced car travel × EPA carbon price.

```
environment = excess_car_VMT
            × 0.000404      (metric tons CO2/mile, EPA)
            × 190           (EPA social cost of carbon, $/ton)
```

- $\text{excess\_car\_VMT} = \text{drive-alone trips} \times \text{home-work distance (LODES)}$
- Fields: ACS B08301 + LODES distance

### 3.4 Education Access

Method: transit deserts → lower attendance → lower lifetime income (Chetty; Minneapolis Student Pass).

```
education = school_age_pop_in_low_access_tract
            × attendance_loss_factor
            × discounted_lifetime_earnings_impact
```

- Fields: ACS B01001, B14001, B15003 + Opportunity Insights coefficient

### 3.5 Forgone Affordability

Method: bad transit forces car ownership; combined housing+transport cost exceeds 45% threshold (H+T Index).

```
forgone_affordability = households_above_45%_HT_threshold
                        × (current_combined_cost - affordable_threshold)
```

- Source: H+T Index + ACS B25064, B25070

### 3.6 Compounding (why it's "cost of doing nothing")

```
Cost(year N) = Σ(components) × (1 + growth_rate)^N
```

Growth from population (Census), wage inflation (BLS), congestion. Computed for N = 1, 3, 5.

### Overlays (not summed into dollars — planner warnings)

- **Safety score:** FARS crash density per transit-dependent tract → flags unsafe walk-to-stop areas.
- **Displacement risk:** income growth + rent burden + demographic vulnerability → flags tracts where investment may price out residents. Pair with affordable housing policy.

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## 4. What the ML Does

### Option A (build first — reliable, explainable):

1. **K-means clustering** — group tracts into vulnerability segments (e.g. "high-need transit desert," "moderate," "well-served") from income, vehicle access, commute time, job access. Gives the planner clean categories instead of raw numbers, and labels who bears the cost.
2. **Time-series forecast (Prophet or linear trend)** — project each cost component forward 1/3/5 years instead of assuming a flat rate. This is what makes the

compounding credible.

**Option B (add only if time permits — ambitious): 3. Regression model** — predict tract-level outcome (e.g. unemployment or poverty rate) as a function of transit access, to estimate impact of investment. Higher risk, needs validation; keep as stretch.

ML role in one line: **clustering says *who is hurt*; forecasting says *how much worse it gets if we wait*.**

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## 5. Tools / Stack

| Purpose        | Tool                                                   |
|----------------|--------------------------------------------------------|
| Language       | Python 3                                               |
| App / UI       | Streamlit                                              |
| Data wrangling | pandas, numpy                                          |
| Geospatial     | geopandas, shapely                                     |
| Maps in app    | pydeck or folium                                       |
| ML             | scikit-learn (K-means, regression), Prophet (forecast) |
| Transit data   | gtfs functions / partridge for GTFS parsing            |
| Charts         | plotly or altair                                       |
| Census pull    | census API (cenpy or requests)                         |

Cache all pulled data to `/data` as CSV/parquet. The live app reads local files only.

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## 6. Dashboard Design (Streamlit)

**Left sidebar — controls (the human-in-the-loop):**

- City: Columbus (fixed for v1)
- Investment amount (\$ slider)
- Delay scenario: Now / 3 yr / 5 yr
- Target tracts (multi-select or map click)
- Assumption sliders: missed-care rate, cost per episode

## Main panel — 4 tabs:

1. **Map** — Columbus tracts colored by Transit Hardship Index. Click a tract → its cost breakdown. Toggle overlays: safety, displacement risk.
2. **Cost of Inaction** (headline tab) — big number at top: "Delaying 5 years costs \$X." Below: stacked timeline chart showing the five components compounding across years 1→5. This is the screenshot for the pitch.
3. **Equity** — who bears the cost. Bar charts by income / race / vehicle access, using K-means segments. Names the disproportionately affected groups.
4. **Scenario Compare** — side by side: "Invest \$X now" vs "Wait 5 years." Shows cost avoided, plus displacement risk warning. Optional: benchmark line as "what good looks like" — use Melbourne (free CBD trams, frequent rail; a car-owning Western city that still built usable transit, so it beats the "only works in dense Asian megacities" objection).

**Footer:** Export → one-page policy brief (PDF) with the numbers and assumptions.

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## 7. Responsible AI (judges weight this)

- **Human decides, not AI.** Tool surfaces costs and tradeoffs; planner makes the funding call. State this explicitly.
  - **One named risk + mitigation:** False positives — a tract flagged "high-need" could misdirect funds. Mitigation: show confidence/data quality per tract, require planner review, never auto-allocate.
  - **Visible assumptions.** Every slider (missed-care rate, cost per episode, growth rate) is exposed and adjustable, not hidden.
  - **Displacement honesty.** Tool warns that investment can price out the residents it aims to help — a tradeoff, not a pure win.
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## 8. Pitch Framing (carry into the video)

The headline of "cost of doing nothing" isn't a dollar figure — it's a person. Lead with one, then back it with the numbers.

- The strongest cases are the transit-dependent: someone who can't drive, an older adult who's aged out, a parent who can't get a child to the ER and can't afford repeated rideshares.

- Concrete contrast that lands fast: in Melbourne a person can cross the city without a car; in a US sprawl metro a 2-mile trip without a car can mean a 45-minute wait. That gap is exactly what this tool measures — in dollars, by neighborhood.
  - Then show the model: who's affected, how much it costs, how much worse it gets if the city waits.
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## 9. Build Priority

**v1 (must ship):** Data layer → Hardship Index → 4 cost components (wages, healthcare, environment, affordability) → K-means → Streamlit map + cost timeline + scenario compare.

**v1.5 (if on track):** Education component, time-series forecast, safety overlay, displacement overlay, export brief.

**v2 (stretch):** Regression outcome model (Option B), Melbourne benchmark line.

Ship v1 end-to-end first. Add layers only once the core loop works.